

Data Science Fundamentals [SC3021]

Data Science Project Introduction

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DATA SCIENCE IN A NUTSHELL (...FULL STORY IN THE LECTURE)

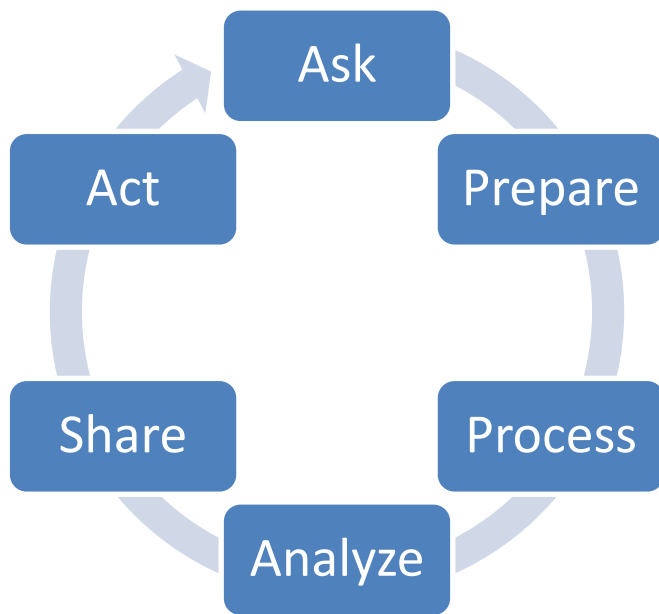


Data Science

- Driven by **domain-specific questions** to be solved based on data
- Requires **several stages** of processing the data (e.g., data collection, analysis, visualization)
- Relies on **methods from different but related disciplines** chosen based on the available data and problem to be solved.
- **No one size fits all** solution, careful engineering throughout the whole process.
- Devising a solution follows a **common process** you will learn more about throughout the course.



Data Science Process



COURSE PROJECT



Objectives

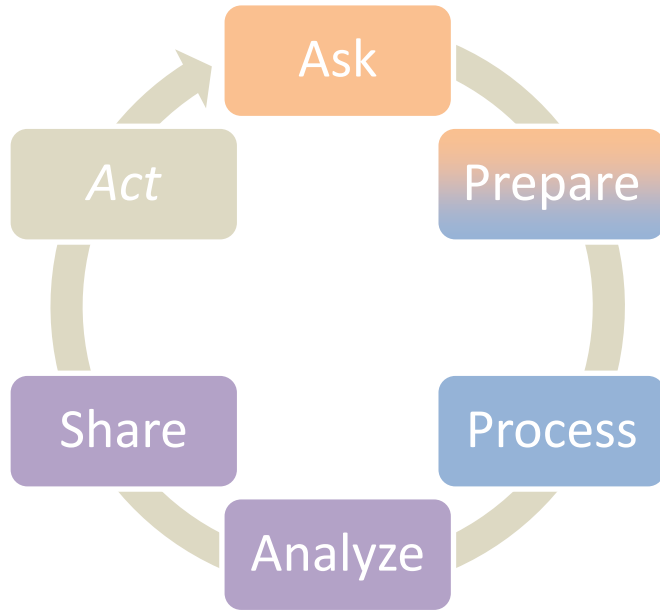
- The goal of the project is to apply the foundational knowledge gained in class by devising and implementing an **end-to-end solution to a data science problem of your choice**.
- Upon completion of the assignment, you should be able to:
 - Use **data science thinking** to address a specific data science problem.
 - **Design a data science solution** for a specific data science problem by identifying and employing appropriate data preparation, data storage and management, modelling and analytics, and data visualization techniques.
 - **Implement the various components** of the designed data science ecosystem using Python.
 - **Reflect** on the developed data science application and potential issues with respect to data protection, ethics, social, and policy aspects.

Project and Lab Manual

- Full manual available in NTULearn
- Short overview of main points here, refer to the manual for further details.
- It is expected that students carefully read and understand the manual.
- Any open questions can be directed to lab TA (= lab supervisor) or instructor
→ may lead to updates to manual for continuous improvement of course material.



What the project covers, and when



Implemented and documented as part of Deliverable 1

Implemented and documented as part of Deliverable 2

Implemented and documented as part of Deliverable 3

Discussion and reflection as part of Deliverable 3

Project deliverables

Deliverable	Deadline	Deliverable	Content	Assessment
Deliverable 1	1 st lab session - 2 days	Executable and documented notebook Slides	Completed problem formulation, requirement analysis, and source identification	Formative
Deliverable 2	2 nd lab session - 2 days	Executable and documented notebook Slides	Completed justified source selection and data preparation	
Deliverable 3	Friday of Week 13	Finalized notebook (code, comments, documentation, discussion, reflection) Video	<u>Notebook</u> : All data science steps and documentation following the guidelines. <u>Video</u> : 3-5 min run-through of notebook with audio, focusing on main highlights/challenges of the project <u>Appendices</u> : peer assessment and AI declaration	Graded

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Deliverable 2	2 nd lab session - 2 days	Executable and documented notebook Slides	Completed justified source selection and data preparation	
Deliverable 3	Friday of Week 7	Finalized notebook (code, comments, documentation, discussion, reflection) Video	<u>Notebook</u> : All data science steps and documentation following the guidelines. 2-3 min run-through of notebook with audio, focusing on main highlights/elements of the project <u>Appendix</u> : Declaration of work	Graded

To be presented by all team members during Lab session 1

Attendance is mandatory

To be presented by all team members during Lab session 2

Deliverables in Python Notebook

- Use of Google **Colab Notebooks** (need to create a **Google account** or use existing one)
 - <https://colab.research.google.com/>
 - <https://www.youtube.com/watch?v=inN8seMm7UI>
- Should follow **template and guidelines** provided by the exemplar notebook
 - <https://colab.research.google.com/drive/1AUkXZBoMH5isArLiQeQ4mfMunxbXTL46?usp=sharing>



PRESENTATION OF GUIDELINES AND EXEMPLAR



TEAM FORMATION (2-3 STUDENTS PER TEAM)



Form teams

- 2-3 students per team – no exceptions
- Reach out to fellow students from the same Lab group to form groups.
- Once you found a group, all members can self-enrol to a (free) group in the Lab NTU Folder.
- To leave a group or change to another group, please reach out to your Lab TA.
- First deliverable is due in Week 4, so the earlier you find a group to get started, the better.

WOOCCLAP ACTIVITY



Recap Quiz

Call for participation

- Each student individually fills the quiz after studying the guidelines.
- Allows us to verify that important aspects of the project are clear or identify points to be clarified.
- Answers submitted in time for lecture preparation will form the basis for the lecture Q&A.

Wooclap

- Link to self-paced Wooclap quiz:
<https://app.wooclap.com/HSNLNS/questionnaires/6926cce283d0ef232b4057e1>
- Login using your NTU SSO.